

AMENDMENT NO. 1 SEPTEMBER 2017
TO
IS 13920 : 2016 DUCTILE DESIGN AND DETAILING OF REINFORCED —
CONCRETE STRUCTURES SUBJECTED TO SEISMIC FORCES — CODE OF
PRACTICE

(First Revision)

[Page 1, clause 2, IS 1893 (Part 1)] — Substitute the following for the existing entry:

‘(Part 1) : 2016 General provisions and buildings (*sixth revision*)’

(Page 2, clause 3.8.1, line 4) — Substitute ‘7.6’ for ‘8’.

(Page 4, clause 5.5, last sentence) — Substitute the following for the existing sentence:

‘When irregularities appear in a building as listed in Tables 5 or 6 of IS 1893 (Part 1), the guidance given therein for the respective irregularity in Tables 5 or 6 shall be followed.’

[Page 5, clause 6.2.6.2 (a)] — Substitute the following for the existing:

‘a) only those mechanical couplers which are conforming to IS 16172 and capable of developing the specified tensile strength in spliced bar shall be permitted. At any section, not more than 50 percent of bars shall be coupled, and the next location of couplers shall be at least 300 mm away; and’

[(Page 6, clause 6.3.1(a), lines 4 and 5)] — Substitute ‘extension of 8 times diameter (but not less than 75 mm)’ for ‘extension of 6 times diameter (but not less than 65 mm)’.

[Page 6, Fig. 4(A) and Fig. 4(B)] — Substitute ‘ $8d \geq 75 \text{ mm}$ ’ for ‘ $6d \geq 65 \text{ mm}$ ’.

(Page 7, Fig. 6) — Substitute ‘HOOP SPACING $\leq d/4$, $6 d_b$ and 100 mm’ for ‘HOOP SPACING $\leq d/4$, $8 d_b$ and 100 mm’.

[Page 7, clause 6.3.5 (b)] — Substitute the following for the existing:

‘b) 6 times the diameter of the smallest longitudinal bar; and’

(Page 7, clause 6.3.5.2) — Insert the following new clause at the end:

‘6.3.5.3 Construction joint shall not be provided in the regions of beam having closely spaced transverse reinforcements.’

(Page 7, clause 7.1, para 2) — Substitute the following for the existing para:

‘For columns of shapes other than rectangular and circular (such as ‘T’, ‘X’ and ‘+’ shaped), which form part of the lateral load resisting system, appropriate designs and detailing shall be carried out using specialist literature.’

(Page 7, Fig. 7) — Substitute ‘ $\geq 20d_b$ ’ for ‘ $\geq 15d_b$ ’ in both figures.

(Page 7, clause 7.1.2, line 4) — Substitute ‘0.4’ for ‘0.45’.

(Page 8, clause 7.2.1.1, second sentence) — Substitute the following for the existing sentence:

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‘The design moment of resistance of a column shall be estimated for factored axial forces arising in all load combination using the design *P-M* interaction diagram.’

(Page 9, clause 7.3.2.2, second sentence) — Substitute the following for the existing sentence:

‘Further only those mechanical splices capable of developing the specified tensile strength of spliced bar shall be permitted. At any section, not more than 50 percent of bars shall be coupled. The next section of couplers shall be at least 300 mm away.’

(Page 9, clause 7.3.2.3, line 4) — Substitute ‘section,’ for ‘location’.

(Page 9, clause 7.4.1, lines 6 and 7) — Substitute ‘8 times’ for ‘6 times’ and ‘< 75 mm’ for ‘< 65 mm’.

[Page 9, clause 7.4.2 (c), line 2] — Insert ‘(h'_c or B'_c)’ after the word ‘link’.

[Page 9, clause 7.4.2 (d), line 4] — Substitute ‘7.6’ for ‘8’.

[Page 9, clause 7.4.2 (d)] — Insert the following at the end:

‘e) Construction joints shall not be provided in regions of columns with closely spaced transverse reinforcement.’

(Page 10, Fig. 10) — Substitute the following for the existing figure:

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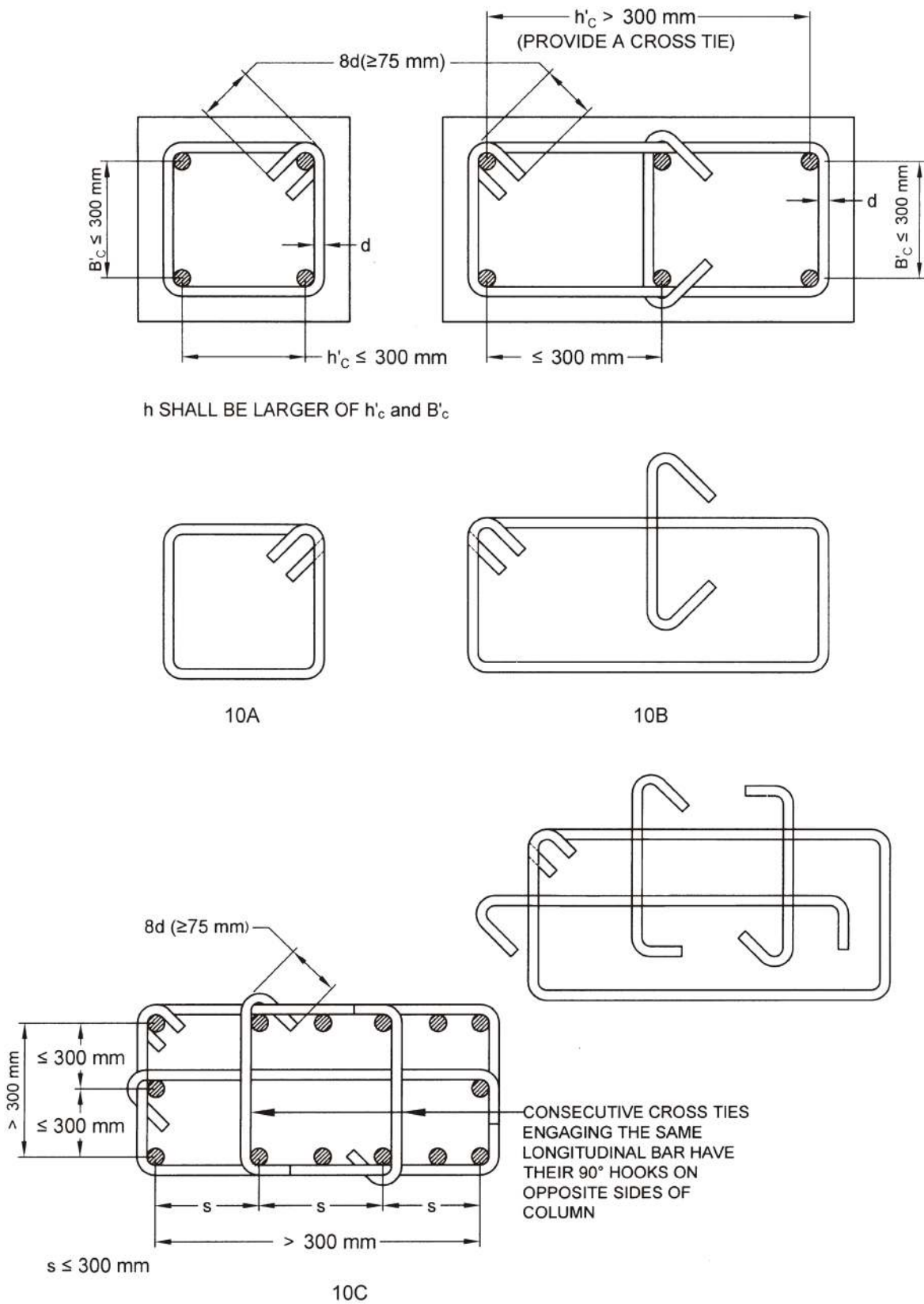


FIG. 10 DETAIL OF TRANSVERSE REINFORCEMENT IN COLUMNS

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(Page 11, clause 8) — Substitute ‘7.6’ for ‘8’.

(Page 11, clause 8.1, line 1) — Renumber the clause as: ‘7.6.1’; and in line 1, delete the words ‘in beams’ and in line 2, delete the word ‘and’.

[Page 11, clause 8.1(b)] — Substitute the following for the existing:

‘b) have a spacing not more than 6 times the diameter of the smallest longitudinal reinforcement bars.’

(Page 11, Fig. 11) — Substitute the following for the existing figure:

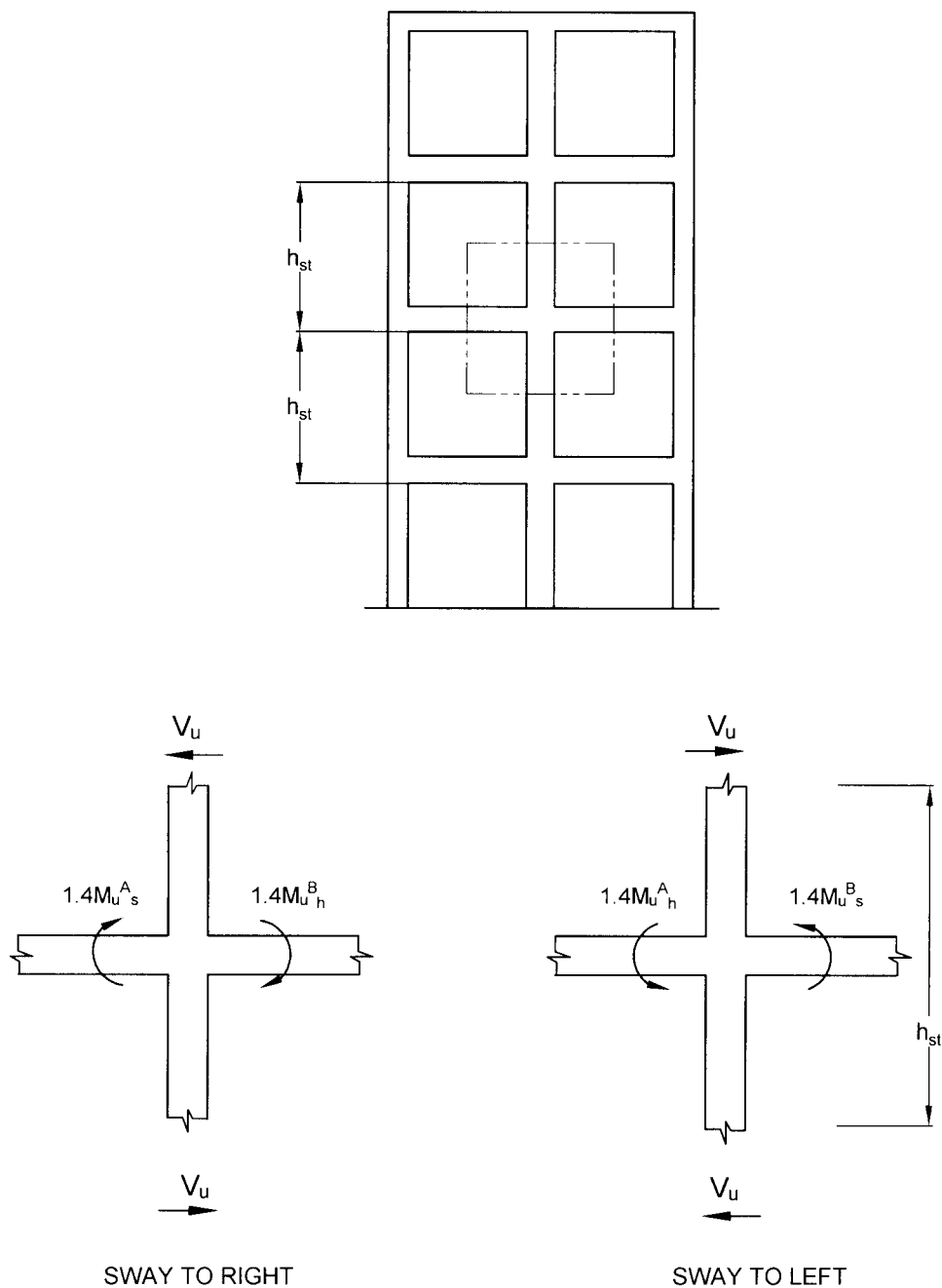


FIG. 11 EQUILIBRIUM DESIGN SHEAR FORCE DEMAND ON COLUMN WHEN PLASTIC HINGES ARE FORMED AT BEAM ENDS

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(Page 12, Fig. 12) — Substitute the following for the existing figure:

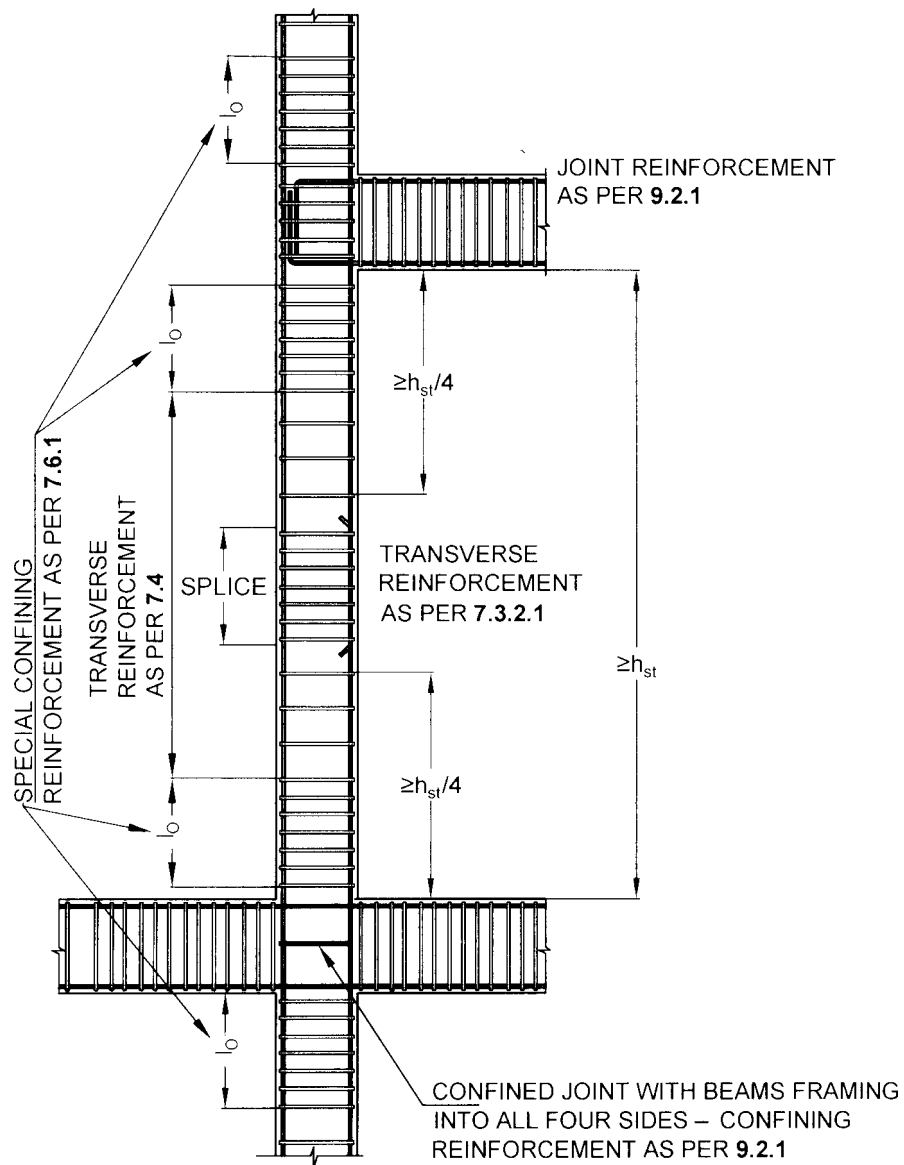


FIG. 12 COLUMNS AND JOINT DETAILING

(Page 12, clause 8.2) — Renumber the clause as '7.6.2'.

(Page 12, clause 8.3) — Renumber the clause as '7.6.3'.

(Page 12, clause 8.4) — Renumber the clause as '7.6.4'.

(Page 13, clause 8.5) — Renumber the clause as '7.6.5'.

(Page 13, clause 9.1.3, line 2) — Insert 'and the shear wall' before the word 'parallel'.

[Page 14, clauses 9.2.1 (a) (1) and 9.2.1 (b) (2), line 2] — Substitute '7.6' for '8'.

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[Page 15, Table 1, Sl No. (i), col 3] — Substitute ' L_w ' for ' t_w ' in the formula for $(\rho_v)_{\min}$.

[Page 15, Table 1, Sl No. (iii), col 3] — Substitute ' $\rho_{v,web}$ ' for ' ρ_h ' in the formula for $(\rho_h)_{\min}$.

(Page 16, clause 10.4.5, line 3) — Substitute '**7.6**' for '**8**'.

[Page 16, clause 10.5.2 (b), line 5] — Substitute '**7.6**' for '**8**'.

(Page 18, clause 11.1.2, line 4) — Substitute '**7.6**' for '**8**'.

(Page 18, clause 11.2.2, line 1) — Substitute '**7.6**' for '**8**'.

[Third cover page, Foreword, Sl No. (c) (6)] — Delete.

[Third cover page, Foreword, Sl No. (c) (7)] — Substitute '**7.6.1(c)**' for '**8.1(c)**'.